آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Teaching and Learning Resources for Grade XI Physics

Recommended Key Textbook:

Model Textbook of Physics 11, National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Additional Recommended Resource Material				
Topic No.	**Topic Title**	**Reference Book**	**Topic in the Reference Book**	**Website/ Link**
1	Physical Quantities and Measurement	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 1: Models, Measurements, and Vectors	https://ebooks.stbb.edu.pk/english/class-xi
		Richard-Woodside, Cambridge-*International AS and A-Level Physics*-Revision-Guide-Second-Edition (2015).	Unit 1: Physical Quantities and Units Unit 2: Measurement Techniques	
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 1: Physics and Measurements	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
2	Scalars and Vectors	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 1: Models, Measurements, and Vectors	http://phet.colorado.edu/ https://www.geogebra.org/m/d6n6rk9N https://www.physicsclassroom.com/Physics-Interactives https://ebooks.stbb.edu.pk/english/class-xi
		Schaum's Outline, Murray R. Spiegel, *Vector Analysis*: McGraw-Hill	Unit 1: Vectors and Scalars Unit 2: The Dot and Cross Product	
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 2: Kinematics	
3	Motion in One dimension	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 2: Motion along a Straight Line Unit 7: Work and Energy Unit 8: Momentum	http://phet.colorado.edu/ https://www.ck12.org/book/ck-12-physics-intermediate/ https://www.physicsclassroom.com/Physics-Interactives https://www.walter-fendt.de/html5/phen/ https://www.vascak.cz/?id=1&language=en#kapitola14 https://ebooks.stbb.edu.pk/english/class-xi
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 1: Dynamics	
		Paul G. Hewitt, *Conceptual Physics* Twelfth Edition (2015): Pearson	Unit 3: Linear Motion Unit 6: Momentum Unit 7: Energy	
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 2: Kinematics Unit 5: Work, Energy and Power	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
4	Motion in Two Dimensions	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 6: Circular Motion and Gravitation Unit 9: Rotational Motion	http://phet.colorado.edu/ https://www.ophysics.com/r9.html https://www.walter-fendt.de/html5/phen/ https://www.physicsclassroom.com/Physics-Interactives https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 4: Mechanics and Fluids	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 2: Accelerated Motion Unit 6: Momentum	
5	Deformation of Solids	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 11: Elasticity and Periodic Motion	http://phet.colorado.edu/ https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 6: Elasticity, Molecular Forces, Solid Materials	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 7: Matter and Materials	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
6	Fluids Dynamics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 13: Fluid Mechanics	https://www.walter-fendt.de/html5/phen/ https://www.geogebra.org/m/HR8CzHTy https://www.physicsclassroom.com/Physics-Interactives https://www.vascak.cz/?id=1&language=en#kapitola14 https://ebooks.stbb.edu.pk/english/class-xi
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 16: Fluid Dynamics	
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 6: Fluid Statics Unit 7: Fluid Dynamics	
7	Oscillations	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 11: Periodic Motion	https://ophysics.com/w4.html https://www.walter-fendt.de/html5/phen/ http://phet.colorado.edu/ https://www.physicsclassroom.com/Physics-Interactives https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 17: Oscillations and Waves	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 19: Oscillations	

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				https://ebooks.stbb.edu.pk/english/class-xi
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 11: Oscillations	

Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
8	Waves	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 21: Mechanical Waves Unit 22: Reflections and Normal Modes Unit 23: Sound	http://phet.colorado.edu/ https://www.walter-fendt.de/html5/phen/ https://ophysics.com/w4.html https://www.physicsclassroom.com/Physics-Interactives https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 17: Oscillations and Waves Unit 22: Applications of Sound Waves Unit 23: Waves in Strings and Pipes	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 13: Waves Unit 15: Stationary Waves	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
9	Physical Optics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 36: The Nature and Propagation of Light Unit 39: Interference and Diffraction	http://phet.colorado.edu/ https://www.walter-fendt.de/html5/phen/
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 15: Geometrical Optics Unit 19: Interference of Light Waves Unit 20: Diffraction of Light Waves Unit 21: Polarisation of Light Waves	https://ophysics.com/l5.html https://www.physicsclassroom.com/Physics-Interactives https://www.vascak.cz/?id=1&language=en#kapitola14
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 41: Interference Unit 42: Diffraction Unit 43: Gratings and Spectra Unit 44: Polarisation Unit 45: The Nature of Light	https://ebooks.stbb.edu.pk/english/class-xi
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 13: Physical Optics	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
10	Thermodynamics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 18: The First Law of Thermodynamics Unit 19: The Second Law of Thermodynamics Unit 20: Molecular Properties of Matter	https://www.walter-fendt.de/html5/phen/ https://learncheme.com/simulations/thermodynamics/thermo-1/ https://learncheme.com/quiz-yourself/interactive-self-study-modules/carnot-cycle/carnot-cycle-simulations/ https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 25: Heat and Energy. Heat Capacity, Latent Heat Unit 26: Gas Laws, Basics of Thermodynamics Unit 27: Kinetic Theory of Gases	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 23: The First Law of Thermodynamics Unit 24: Entropy and the Second Law of Thermodynamics	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 21: Thermal Physics Unit 22: Ideal Gas	

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Note: This resource list has been prepared primarily for teachers. While it can be shared with students, students should not be required to buy multiple books. Schools are encourages to stock these books in the library. Moreover, these are only suggestions which have been compiled for the ease of teachers and students; schools are encouraged to use other resources for teaching and learning as well, as long as they are in line with the student learning outcomes (SLOs) mentioned in AKU-EB syllabi. Unless specified, AKU-EB does not endorse any of these books or websites. You are advised to use an ad-blocker while accessing the websites. In case any website is not functional for any reason, you may inform us at examination.board@aku.edu for an alternative or search material via any search engine. If you have any query, please contact us via email.

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Teaching and Learning Resources for Grade XII Physics

Recommended Key Textbook:

Model Textbook of Physics 12, National Book Foundation, Islamabad (Based on National Curriculum of Pakistan 2022-2023)

Additional Recommended Resource Material				
Topic No.	**Topic Title**	**Reference Book**	**Topic in the Reference Book**	**Website/ Link**
11	Gravitation	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 6: Circular Motion and Gravitation	http://phet.colorado.edu/ https://www.physicsclassroom.com/Physics-Interactives https://www.walter-fendt.de/html5/phen/ https://www.vascak.cz/?id=1&language=en#kapitola14
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 18: Gravitational Fields	
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 7: The Gravitational Field	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
12	Electrostatics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 24: Coulomb's Law Unit 25: The Electric Field Unit 26: Electrical Potential Unit 27: Capacitance and Dielectrics	http://phet.colorado.edu/ https://www.physicsclassroom.com/Physics-Interactives https://ophysics.com/l5.html https://www.vascak.cz/?id=1&language=en#kapitola14 https://ebooks.stbb.edu.pk/english/class-xi
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 25: Electric Charge and Coulomb's Law Unit 26: The Electric Field Unit 27: Gauss' Law Unit 28: Electric Potential Energy and Potential Unit 30: Capacitance	
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 8: Electric Field Unit 9: Capacitors	
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 8: Electrostatics. The Electric Field Unit 9: Capacitors	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
13	Electric Current	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 28: Current, Resistance and Electromotive Force Unit 29: Direct-Current Circuits	http://phet.colorado.edu/ https://www.physicsclassroom.com/Physics-Interactives https://www.walter-fendt.de/html5/phen/ https://ophysics.com/l5.html https://www.vascak.cz/?id=1&language=en#kapitola14 https://ebooks.stbb.edu.pk/english/class-xi
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 10: Current Electricity	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 31: DC Circuits	
		Sindh Textbook Board. *The Textbook of Physics for Class XI*. Jamshoro: Sindh Textbook Board.	Unit 10: Current Electricity	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 9: Electric current, potential difference and resistance Unit 10: Kirchhoff's laws Unit 11: Resistance and resistivity Unit 12: Practical circuits	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
14	Electromagnetism	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 30: Magnetic Field and Magnetic Forces Unit 31: Sources of Magnetic Field	http://phet.colorado.edu/ https://www.physicsclassroom.com/Physics-Interactives https://www.walter-fendt.de/html5/phen/ https://ophysics.com/l5.html https://www.vascak.cz/?id=1&language=en#kapitola14 https://ebooks.stbb.edu.pk/english/class-xii
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 11: Magnetic Field and Force on Conductor Unit 12: Magnetic Fields of Current-Carrying Conductors Unit 30: Electron Motion in Fields, Cathode-Ray Oscilloscope	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 32: The Magnetic Field Unit 33: The Magnetic Field of a Current	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 26: Magnetic fields and electromagnetism Unit 27: Charge Particles	
		Sindh Textbook Board. *A Textbook of Physics for Class XII*. Jamshoro: Sindh Textbook Board.	Unit 14: Electromagnetism	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
15	Electromagnetic Induction and Alternating Current	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 32: Electromagnetic Induction Unit 34: Alternating Current	http://phet.colorado.edu/ https://www.walter-fendt.de/html5/phen/ https://ophysics.com/l5.html https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 13: Electromagnetic Induction Unit 14: A.C. Circuits	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 34: Faraday's Law of Induction	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 28: Electromagnetic Induction Unit 29: Alternating Current	
16	Quantum Mechanics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 40: Relativistic Mechanics Unit 42: Quantum Mechanics	http://phet.colorado.edu/ https://www.walter-fendt.de/html5/phen/ https://ophysics.com/l5.html https://www.physicsclassroom.com/Physics-Interactives
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 33: Photoelectricity, Energy Levels, X-Rays, Wave-Particle Duality	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second*	Unit 30: Quantum Physics	

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		Edition (2014): Cambridge University Press.		https://www.vascak.cz/data/android/physicsatschool/template.php?f=str_dilatace&l=en

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
17	Nuclear Physics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 44: Nuclear and High-Energy Physics	http://phet.colorado.edu/ https://www.walter-fendt.de/html5/phen/ https://spark.iop.org/teaching-radioactivity-animations https://www.slideshare.net/slideshow/nuclear-fission-fusion-ppt-animation/127676224 https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 34: Radioactivity, Nuclear Energy	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 50: Nuclear Physics Unit 51: Energy from the Nucleus	
		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 16: Radioactivity Unit 31: Nuclear physics	
18	Particle Physics	Sears, Zamansky and Young Addison, *College Physics* (2011). New York: Wesley Publication Co.	Unit 44: Nuclear and High energy physics	https://www.sciencephoto.com/media/727977/view/subatomic-nuclear-forces-animation http://phet.colorado.edu/ https://www.vascak.cz/?id=1&language=en#kapitola14
		Nelkon and Parker. (Seventh edition) (1995). *Principles of Physics*. Heinemann: Collins Int'l Textbooks.	Unit 34: Radioactivity and nuclear energy	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 52: Particle Physics and Cosmology	

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		David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 16: Radioactivity	

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Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
19	Medical and Space Physics	David Sang, Graham Jones, Gurinder Chadha and Richard Woodside, *Cambridge International AS and A Level Physics Coursebook Second Edition* (2014): Cambridge University Press.	Unit 32: Medical imaging	
		Halliday and Resnick, *Fundamentals of Physics*: Walker Wiley India Pvt. Ltd. (Volume 1) (2007).	Unit 52: Particle Physics and Cosmology	
20	Nature of Science	-	-	https://www.britannica.com/science/scientific-method

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